

Zhou Wei

Gender: Male Objective: Senior Backend Engineer Location: Beijing

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Education

2013.09 - 2017.06 Chongqing University Computer Science and Technology (B.S.)

Courses: Data Structures and Algorithms, Distributed Systems, Machine Learning, Computer Networks, Design Patterns, Compilers

GPA 3.4/4.0, top 30% in major; merit scholarship

Summary

Seven years of backend experience focused on payment/transaction systems, distributed architecture, high concurrency and production incident response.

Work Experience

2020.07 - Present Kuaishou Senior Backend Engineer

- Fixed a log-collection failure caused by agent backpressure, restoring stability after tuning batch and buffer settings.
- Tuned Redis caching for hot keys and cache breakdown with local cache and distributed locks, raising hit rate above 90%.
- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 31%, improving end-to-end delivery quality.
- Optimized a file-sort slow SQL on a hot API, added a composite index via gh-ost online DDL, reducing P99 from 1500ms to 320ms.
- Investigated rising instance memory rss using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 31%.

2017.07 - 2020.06 JD.com Backend Engineer

- Owned backend of core transaction and order services, splitting the monolith into order, settlement and risk-control microservices.
- Served 1270K requests per day while keeping core-path SLA above 99.9%.
- Built service observability covering API latency, error rate, JVM GC, thread pools and consumer backlog with Prometheus and Grafana, and it became a reusable team best practice.
- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism, ensuring online service stability.

Projects

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Used sharding and read-write splitting to keep single-table rows within tens of millions.
- Applied idempotency and eventual consistency to avoid duplicate charges on retry.
- Recorded an audit trace id for every payment state transition for compliance.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction.

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)

- Stayed stable at 30000 QPS in load tests with error rate below 0.1%.
- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes, ensuring online service stability.
- Designed distributed locks and token-bucket rate limiting to prevent oversell.
- Built circuit breaking and fallback for downstream failures, recognized by stakeholders and peers.

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Added circuit breaking to isolate faulty downstreams.
- Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, and it became a reusable team best practice.
- Shipped a unified SDK to lower integration cost across teams, and related standards were adopted by the team.
- Integrated OAuth2 and RBAC with multi-tenant isolation.

Highlights

- Led a major technical effort involving MySQL indexing and slow-query tuning, staying stable at 30000 QPS peak
- Built a reusable methodology and docs around Git collaboration and code review, adopted by several teams
- Established metrics and reviews around Kafka/RocketMQ messaging to keep improving delivery quality
- Work around Distributed transactions and consistency delivered about 31% efficiency gains and was reused across the team
- Owned core modules that steadily served 1270K daily visits with 99.9%+ availability

Skills

Core skills: MySQL, Docker and containerization, JVM, Microservices, High Concurrency

Proficient in: Observability, Distributed transactions and consistency, SQL Optimization, MySQL indexing and slow-query tuning, Redis caching and distributed locks

Familiar with: Kafka/RocketMQ messaging, Spring Boot, Java, Kafka, JVM tuning and troubleshooting, Redis

Self Evaluation

- Solid engineering fundamentals and strong troubleshooting skills, able to own a module end to end
- Business-minded, good at balancing technical design against delivery timelines